

# Raghav Malik

PHD CANDIDATE · COMPUTER ENGINEERING

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## Education

### Purdue University

PHD COMPUTER ENGINEERING

• Advisor: Dr. Milind Kulkarni

BS COMPUTER ENGINEERING

West Lafayette

August 2019 - present

August 2016 - May 2019

## Professional Experience

### Research Assistant, Parallelism, Languages, and Compilers Lab

2019-2024

- Developed a novel abstraction for expressing homomorphic circuits based on programmable bootstrapping
- Designed a protocol for evaluating computations with ciphertext-dependent branches
- Extended general vectorization techniques to apply to encrypted arithmetic circuits
- Developed an algorithm for efficiently vectorizing private decision tree inference
- Contributed to the design of a metaprogramming language for writing cryptographic applications

### Research Intern, Microsoft Research:

2022

- Developed scheduling optimizations for the MSCCL GPU compiler
- Designed an event-based simulation engine for profiling MSCCL communication collectives

### Software Intern, Amazon Robotics:

2021

- Designed a DSL for expressing complex data validation rules

### Software Intern, AWS Kinesis:

2019-2020

- Developed features for Kinesis Firehose for monitoring and streaming to S3

## Publications

**Malik, Raghav**, Vedant Paranjape, Milind Kulkarni. 2025. COATL: Circuit Optimization using Arithmetic Table Lookup. <https://raghav198.github.io/assets/papers/coatl.pdf> [Accepted to PLDI 2025]

**Malik, Raghav**, Kabir Sheth, Milind Kulkarni. 2023. Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits. Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3: 118-133.

**Malik, Raghav**, Vidush Singhal, Benjamin Gottfried, Milind Kulkarni. 2021. Vectorized Secure Evaluation of Decision Forests. Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation: 1049-1063.

Bao, Yuyan, Kirshanthan Sundarajah, **Raghav Malik**, Qianchuan Ye, et. al. 2021. HACCLE: Metaprogramming for Secure Multi-Party Computation. Proceedings of the 20th ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences: 130-143.

## PREPRINTS

**Malik, Raghav**, Milind Kulkarni. 2024. COIL: Compiling Homomorphic Circuits with Control Flow. <https://raghav198.github.io/assets/papers/coil.pdf>

Sreekanth, Sreevickrant, Dulani Wijayarathne, **Raghav Malik**, Milind Kulkarni. 2024. Biscotti: An Approach to Parallel Scheduling for Vectorized Encrypted Arithmetic Circuits. <https://raghav198.github.io/assets/papers/biscotti.pdf>

## Presentations

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\* *presenting author*; + *mentored undergraduate*

### INVITED TALKS

Fall 2024. *Compiling Homomorphic Circuits with Control Flow*. Poster: Midwestern PL Summit 2024

Fall 2023. *Managing Ownership and Lifetimes*. Guest lecture, Object Oriented Programming in C++, Purdue University

Fall 2023. *Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits*. Invited talk: Midwestern PL Summit 2023

Spring 2023. *Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits*. Invited talk: Galois, Inc.

Fall 2022. *Vectorized Secure Evaluation of Decision Forests*. Invited talk: Cornell University.

### CONTRIBUTED PRESENTATIONS

Sreekanth, Vickranth<sup>+</sup>, Dulani Wijayarathne<sup>+</sup>, **Raghav Malik**, Milind Kulkarni. 2023. Parallel Encrypted Arithmetic Vector Scheduling for Optimized FHE Computations. Departmental seminar: Purdue Programming Languages Group.

Sreekanth, Vickranth<sup>+</sup>, Dulani Wijayarathne<sup>+</sup>, **Raghav Malik**, Milind Kulkarni. 2023. An Optimization for Vectorizing Encrypted Arithmetic Circuits. Poster: Summer Undergraduate Research Symposium, Purdue University.

Bao, Yuyan, Kirshanthan Sundarajah<sup>\*</sup>, **Raghav Malik**, Qianchuan Ye, et. al. 2021. HACCLE: Metaprogramming for Secure Multiparty Computation. GPCE Conference Talk.

## Projects

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### CATEGORIAL NERVE IN LEAN

Used Lean to define and formalize explicit constructions of the nerve of a category and the homotopy realization of a simplicial set, and proved an adjunction between these functors.

### PROOFMOUSE

A lightweight proof assistant for verifying Gentzen-style propositional and predicate logic proofs. ProofMouse is currently being used as a teaching aid to help discrete mathematics students learn logic and how to write proofs.

## Teaching Experience

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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fall 2024   | <b>Object Oriented Programming in C++</b> , Teaching Assistant<br><i>Designed and delivered some lectures, wrote exams, managed undergraduate TAs</i>       |
| Summer 2024 | <b>Data Structures and Algorithms</b> , Instructor<br><i>Designed an 8-week course, designed and delivered lectures, problem sets, and exams.</i>           |
| Spring 2024 | <b>Object Oriented Programming in C++</b> , Teaching Assistant<br><i>Held office hours, managed undergraduate TAs, designed and delivered some lectures</i> |
| Fall 2023   | <b>Object Oriented Programming in C++</b> , Teaching Assistant<br><i>Held office hours, developed autograding framework, delivered guest lecture</i>        |
| Spring 2023 | <b>Discrete Mathematics</b> , Teaching Assistant<br><i>Held office hours, developed a proof assistant for automatically verifying Gentzen-style proofs</i>  |

## Mentoring

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| 2022-2024 | <b>Dulani Wijayarathne</b> , Summer Undergraduate Research Student |
| 2022-2024 | <b>Vickranth Sreekanth</b> , Summer Undergraduate Research Student |
| 2021-2022 | <b>Kabir Sheth</b> , Undergraduate Research Assistant              |
| 2019-2021 | <b>Vidush Singhal</b> , Undergraduate Research Assistant           |
| 2019-2020 | <b>Benjamin Gottfried</b> , Undergraduate Research Assistant       |

## Skills

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### TECHNICAL SKILLS

C++, Python, OCaml, Haskell, Linux, Git, object-oriented programming, functional programming

### RESEARCH SKILLS

Compiler optimizations, vectorization, cryptography, homomorphic encryption, multiparty computation, program synthesis

### SOFT SKILLS

Mentoring undergraduates, designing and teaching courses, writing grants, collaborating on research

## Professional Development

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### SERVICE

2024-2025 **Association of Computing and Machinery, PLDI**, Organizing Committee, Student Volunteer  
Chair

2022-2023 **Association of Computing and Machinery, PLDI**, Student Volunteer

### PEER REVIEW

Program Committee, ARRAY 2025

Artifact Evaluation Committee, PPOPP 2024